

7.3 Final Lab Practical

Overview

This assignment is the culminating, summative lab assignment for IT 350 – Network Configuration and Design.

In this final lab, you will demonstrate your ability to integrate, configure, verify, and troubleshoot a complete network using the technologies and skills you have developed throughout the course. You are expected to approach this work the way a network technician or junior network engineer would in a real workplace environment.

Files

Download the two files provided for this assignment:

- [IT 350 - Final Lab Practical.docx](#)Download IT 350 - Final Lab Practical.docx
- [IT 350 - Final Lab Practical - Topology.pkt](#)Download IT 350 - Final Lab Practical - Topology.pkt

Scenario

- Please see *IT 350 - Final Lab Practical.docx* for the scenario you will be working with.

Purpose of the Worksheet and Screenshots

- The Final Lab Practical worksheet defines the requirements for this assignment and provides examples of verification evidence, including screenshots, to help you confirm your configuration.
- Screenshots shown in the worksheet illustrate what correct results look like; you are not required to submit screenshots or the worksheet itself.
- Final submission consists of the Packet Tracer (.pkt) file and the show run output for all required devices, as described in the final step of the worksheet.

Instructions

- **Download** the two files provided in this assignment.

- **Follow** the instructions listed in the document, *IT 350 - Final Lab Practical.docx*, to complete your project (to satisfy all requirements and achieve full connectivity between all PCs in the network).
- **Capture** the required "show run" output file(s):
 - Save your show run on your routers and switch with a wr mem and copy them to a text file.
 - You may use one text file for all devices or may upload a separate text file for each device.
 - If you use one file, please clearly label the beginning of each device's output with the device name so that it is clear where each device's show run begins and ends.
 - If you use multiple files, please name them with the hostname of the device.
- Submit the following to Canvas
 - Your completed **Packet Tracer (.pkt)** file
 - The **"show run" output file(s)** for all devices (R1-SITE1, R2-SITE2, S1-SITE1, S2-SITE1, and S3-SITE1)

Grading

Your assignment will be evaluated according to the criteria outlined in the rubric.

Rubric

7.3 Final Lab Practical - Rubric

7.3 Final Lab Practical - Rubric

Criteria	Ratings	Pts
This criterion is linked to a Learning Outcome Device Hostnames	8 ptsFull MarksAll required routers and switches are configured with hostnames that exactly match the topology labels. The submitted evidence (for example, output from the command "show cdp neighbor" captured on S1-SITE1) clearly shows all expected device names with no discrepancies. 7 ptsMost MarksHostnames are correctly configured on most required devices, but one device has a minor hostname issue (such as a typo, formatting difference, or missing hostname). Evidence shows the remaining device hostnames match the topology labels. 5 ptsModerate MarksHostnames are configured on some required devices, but multiple devices have missing or incorrect hostnames. Evidence shows partial completion of the hostname requirement. 3 ptsSome MarksMinimal hostname configuration is present. One or more devices may show a hostname change, but most required devices do not match the topology labels or evidence is insufficient. 0 ptsNo MarksNo evidence is provided showing hostname configuration, or the submitted evidence does not demonstrate that any required device hostnames were configured.	8 pts

7.3 Final Lab Practical - Rubric

Criteria	Ratings	Pts
This criterion is linked to a Learning OutcomeInterface Status (Green Links)	8 ptsFull MarksAll required interfaces show an up/up state, and the submitted evidence (for example, a screenshot of the logical view) clearly shows green links on all devices with no interface issues present. 7 ptsMost MarksMost required interfaces show a green (up/up) status, but one interface is down or incorrectly configured. The submitted evidence otherwise demonstrates that the majority of interfaces are operational. 5 ptsModerate MarksSome interfaces show a green (up/up) status, but multiple required interfaces are down or not correctly configured. The submitted evidence shows partial completion of the interface status requirement. 3 ptsSome MarksMinimal progress toward correct interface status is evident. One or more interfaces may be up, but most required interfaces are down, or the submitted evidence is insufficient to verify interface status. 0 ptsNo MarksNo evidence is provided showing interface status, or the submitted evidence does not demonstrate that any required interfaces are operational.	8 pts
This criterion is linked to a Learning OutcomeIP Addressing Configuration	8 ptsFull MarksAll required routers and switches are configured with the correct IP addressing as specified. Submitted evidence (for example, output from the command "show ip int brief") clearly shows valid IP addresses on all required interfaces with no missing or incorrect configurations. 7 ptsMost MarksIP addressing is correctly configured on most required interfaces, but one interface has a minor addressing issue (such as a missing, incorrect, or misassigned IP address). Evidence shows correct addressing on the remaining interfaces. 5 ptsModerate MarksIP addressing is configured on some required interfaces, but multiple interfaces have missing or	8 pts

7.3 Final Lab Practical - Rubric

Criteria	Ratings	Pts
	incorrect IP addresses. Evidence shows partial completion of the IP addressing requirement. 3 ptsSome MarksMinimal IP addressing configuration is present. One or more interfaces may have an IP address assigned, but most required interfaces are missing or incorrectly configured, or evidence is insufficient to verify addressing. 0 ptsNo MarksNo evidence is provided showing IP addressing configuration, or the evidence does not demonstrate that any required interfaces have been configured with IP addresses.	
This criterion is linked to a Learning OutcomeSecure Device Access Configuration	8 ptsFull MarksAll required devices are configured with secure access settings as specified. Submitted evidence (for example, output from the command "show run section line") clearly shows an enable password, console password, and telnet access configured on all required devices with no missing elements. 7 ptsMost MarksSecure access settings are correctly configured on most required devices, but one device is missing a required access element (such as an enable password, console password, or telnet configuration). Evidence otherwise shows secure access configured on the remaining devices. 5 ptsModerate MarksSome secure access settings are configured, but multiple required devices are missing one or more required access elements. Evidence shows partial completion of the secure device access requirement. 3 ptsSome MarksMinimal secure access configuration is present. One or more access elements may be configured on a limited number of devices, but most required devices are	8 pts

7.3 Final Lab Practical - Rubric

Criteria	Ratings	Pts
	missing secure access settings, or evidence is insufficient to verify configuration. 0 ptsNo MarksNo evidence is provided showing secure device access configuration, or the submitted evidence does not demonstrate that any required devices have secure access settings configured.	
This criterion is linked to a Learning OutcomeVLAN Creation and Trunk Configuration	8 ptsFull MarksThe required VLAN is created on all SITE1 switches, access ports are assigned as specified, and trunk links are configured to carry the required VLANs. Submitted evidence (for example, output from "show vlan" and "show interfaces trunk") demonstrates correct VLAN and trunk configuration across all switches. 7 ptsMost MarksVLAN and trunk configuration is mostly correct, but one switch has a minor issue (such as a missing VLAN, incorrect access port assignment, or trunk configuration problem). Evidence shows correct configuration on the remaining switches. 5 ptsModerate MarksVLAN and trunk configuration is partially complete. The required VLAN exists on some switches or trunking is configured on some links, but multiple required switches or links are missing correct configuration. 3 ptsSome MarksMinimal VLAN or trunk configuration is present. One or more switches may show VLAN or trunk	8 pts

7.3 Final Lab Practical - Rubric

Criteria	Ratings	Pts
	configuration, but most required switches or links are not correctly configured, or evidence is insufficient to verify setup. 0 ptsNo MarksNo evidence is provided showing VLAN creation or trunk configuration, or the submitted evidence does not demonstrate that any required VLAN or trunk settings are configured.	
This criterion is linked to a Learning OutcomeInterVLAN Routing (R1-SITE1)	8 ptsFull MarksInterVLAN routing is correctly configured on R1-SITE1, allowing traffic to route between the required VLANs. Submitted evidence (for example, output from "show run section interface") demonstrates correctly configured subinterfaces with no missing or incorrect elements. 7 ptsMost MarksInterVLAN routing is mostly correct, but one minor issue is present (such as a missing, incorrect, or misconfigured subinterface). Evidence otherwise shows routing configured between the VLANs. 5 ptsModerate MarksInterVLAN routing is partially configured. One or more required subinterfaces are present, but routing between VLANs is incomplete or incorrectly configured. 3 ptsSome MarksMinimal interVLAN routing configuration is present. Evidence shows an attempt to configure routing on R1-SITE1, but required VLAN routing is largely missing or incorrect.	8 pts

7.3 Final Lab Practical - Rubric

Criteria	Ratings	Pts
	0 ptsNo MarksNo evidence is provided showing interVLAN routing configuration on R1-SITE1, or the submitted evidence does not demonstrate any routing between VLANs.	
This criterion is linked to a Learning OutcomeDHCP Configuration	8 ptsFull MarksDHCP is correctly configured as specified, and submitted evidence (for example, output from "show run section ip dhcp") shows a complete DHCP configuration with required pools and exclusions present. 7 ptsMost MarksDHCP is mostly configured correctly, but one minor issue is present (such as a missing exclusion or incomplete pool configuration). Evidence otherwise shows DHCP functioning as intended. 5 ptsModerate MarksDHCP is partially configured. A DHCP pool exists, but required elements are missing or incorrect, and the configuration does not fully meet the specified requirements. 3 ptsSome MarksMinimal DHCP configuration is present. Evidence shows an attempt to configure DHCP, but required pools or settings are largely missing or incorrect.	8 pts

7.3 Final Lab Practical - Rubric

Criteria	Ratings	Pts
	0 ptsNo MarksNo evidence is provided showing DHCP configuration, or the submitted evidence does not demonstrate that DHCP is configured.	
This criterion is linked to a Learning OutcomeNAT / PAT Configuration	8 ptsFull MarksNAT / PAT is correctly configured as specified. Submitted evidence (for example, output from "show run section ip nat" or "show ip nat statistics") demonstrates source NAT with overload configured and operating on the required device. 7 ptsMost MarksNAT / PAT is mostly configured correctly, but one minor issue is present (such as a missing statement or incorrect inside/outside designation). Evidence otherwise shows NAT / PAT configured. 5 ptsModerate MarksNAT / PAT is partially configured. Some required NAT statements or interface designations are present, but the configuration does not fully meet the specified requirements. 3 ptsSome MarksMinimal NAT / PAT configuration is present. Evidence shows an attempt to configure NAT, but required statements or interface roles are largely missing or incorrect. 0 ptsNo MarksNo evidence is provided showing NAT / PAT configuration, or the submitted evidence does not demonstrate that NAT / PAT is configured.	8 pts

7.3 Final Lab Practical - Rubric

Criteria	Ratings	Pts
This criterion is linked to a Learning Outcome Static Default Route Configuration	9 ptsFull MarksA static default route is correctly configured as specified. Submitted evidence (for example, output from "show ip route") clearly shows a default route pointing to the required next hop with no errors. 8 ptsMost MarksA static default route is present, but one minor issue exists (such as an incorrect next hop or incomplete configuration). Evidence otherwise shows a default route configured. 6 ptsModerate MarksA static default route is partially configured. Evidence shows an attempt to configure a default route, but the configuration does not fully meet the specified requirements. 3 ptsSome MarksMinimal default route configuration is present. Evidence shows an incomplete or incorrect default route, or the submitted evidence is insufficient to verify proper configuration. 0 ptsNo MarksNo evidence is provided showing static default route configuration, or the submitted evidence does not demonstrate that a default route is configured.	9 pts
This criterion is linked to a Learning Outcome OSPF Configuration on R2-SITE2	9 ptsFull MarksOSPF is correctly configured on R2-SITE2 as specified. Submitted evidence (for example, output from "show ip ospf interface") clearly shows an active OSPF process with the required interfaces participating and no configuration issues present. 8 ptsMost MarksOSPF is mostly configured correctly on R2-SITE2, but one minor issue is present (such as an incorrect network statement or missing passive interface). Evidence otherwise shows OSPF enabled and operating. 6 ptsModerate MarksOSPF is partially configured on R2-SITE2. Evidence shows an OSPF process present, but required	9 pts

7.3 Final Lab Practical - Rubric

Criteria	Ratings	Pts
	interfaces are missing, misconfigured, or not participating as specified. 3 ptsSome MarksMinimal OSPF configuration is present. Evidence shows an attempt to configure OSPF on R2-SITE2, but the configuration does not meet the required specifications. 0 ptsNo MarksNo evidence is provided showing OSPF configuration on R2-SITE2, or the submitted evidence does not demonstrate that OSPF is configured.	
This criterion is linked to a Learning OutcomeRedundant Links and Channel Groups	9 ptsFull MarksRedundant links and channel groups are correctly configured as specified. Submitted evidence (for example, a logical view or relevant configuration output) clearly shows the required redundant links grouped correctly with no missing or incorrect channel group configuration. 8 ptsMost MarksRedundant links and channel groups are mostly configured correctly, but one minor issue is present (such as a missing link in a channel group or an incorrectly configured group). Evidence otherwise shows redundant links grouped as intended. 6 ptsModerate MarksRedundant links and channel groups are partially configured. Evidence shows some links grouped into channel groups, but multiple required links or groups are missing or incorrectly configured. 3 ptsSome MarksMinimal redundant link or channel group configuration is present. Evidence shows an attempt to configure redundant links, but most required links are not correctly grouped, or evidence is insufficient to verify proper configuration.	9 pts

7.3 Final Lab Practical - Rubric

Criteria	Ratings	Pts
	0 ptsNo MarksNo evidence is provided showing redundant links or channel group configuration, or the submitted evidence does not demonstrate that any required redundant links are configured.	
This criterion is linked to a Learning OutcomeSpanning Tree Configuration	9 ptsFull MarksSpanning Tree is configured as specified. Submitted evidence (for example, output from "show spanning-tree") clearly shows the correct root bridge placement and expected spanning-tree state with no configuration issues present. 8 ptsMost MarksSpanning Tree is mostly configured correctly, but one minor issue is present (such as incorrect root priority on one device). Evidence otherwise shows spanning-tree enabled and operating. 6 ptsModerate MarksSpanning Tree is partially configured. Evidence shows spanning-tree enabled, but the root bridge or spanning-tree state does not fully match the specified requirements. 3 ptsSome MarksMinimal spanning-tree configuration is present. Evidence shows spanning-tree activity, but required root placement or configuration is largely missing or incorrect.	9 pts

7.3 Final Lab Practical - Rubric

Criteria	Ratings	Pts
	0 ptsNo Marks No evidence is provided showing spanning-tree configuration, or the submitted evidence does not demonstrate that spanning-tree is configured.	

Total Points: 100